

Supplemental Table 1. Complete ‘*glmmTMB*’ seasonality model outputs for model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites, model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites, and model 3 measuring hurricane effects on myxozoan parasites. Bolded values represent significance and bolded model terms represent significant values on both short- and long- time scales.

response variable	model term	3 month						6 month					
		estimate	std.error	CI_lower	CI_upper	z-statistic	p.value	estimate	std.error	CI_lower	CI_upper	z-statistic	p.value
direct or complex life cycle strategy	(Intercept)	44.6251	0.1532	33.0514	60.2518	24.7961	0.0000	47.5209	0.1532	35.1949	64.1639	25.2037	0.0000
	seasonspring	1.1653	0.7649	0.2602	5.2178	0.2000	0.8415	0.9712	0.2825	0.5582	1.6896	-0.1036	0.9175
	seasonsummer	0.9029	0.2380	0.5663	1.4397	-0.4289	0.6680	0.9409	0.3225	0.5000	1.7706	-0.1888	0.8503
	seasonwinter	1.1030	0.2931	0.6210	1.9592	0.3345	0.7380	1.0055	0.2145	0.6604	1.5308	0.0254	0.9798
	eventhurricane	0.8598	0.2192	0.5596	1.3212	-0.6892	0.4907	0.8047	0.2043	0.5392	1.2009	-1.0635	0.2875
	seasonspring*eventhurricane	0.9908	0.8498	0.1873	5.2407	-0.0109	0.9913	1.4535	0.3448	0.7394	2.8571	1.0846	0.2781
	seasonsummer*eventhurricane	1.3722	0.3221	0.7299	2.5796	0.9824	0.3259	1.2873	0.3774	0.6144	2.6972	0.6692	0.5033
	seasonwinter*eventhurricane	0.2835	0.8391	0.0547	1.4685	-1.5020	0.1331	0.8950	0.2982	0.4989	1.6056	-0.3720	0.7099
monogenean, trematode adult, trematode metacercaria	(Intercept)	18.2820	0.1065	14.8373	22.5265	27.2810	0.0000	16.2916	0.1046	13.2730	19.9968	26.6918	0.0000
	seasonspring	1.0940	0.3966	0.5028	2.3802	0.2266	0.8208	1.1078	0.1888	0.7652	1.6037	0.5423	0.5876
	seasonsummer	1.0666	0.1722	0.7611	1.4947	0.3746	0.7080	1.0980	0.1994	0.7428	1.6232	0.4690	0.6391
	seasonwinter	1.0970	0.1888	0.7576	1.5884	0.4903	0.6240	1.1867	0.1413	0.8996	1.5654	1.2115	0.2257
	eventhurricane	0.6902	0.1499	0.5145	0.9260	-2.4732	0.0134	0.8210	0.1357	0.6293	1.0710	-1.4541	0.1459
	seasonspring*eventhurricane	0.6760	0.5774	0.2180	2.0963	-0.6781	0.4977	1.2045	0.2243	0.7760	1.8697	0.8296	0.4068
	seasonsummer*eventhurricane	1.3888	0.2282	0.8879	2.1721	1.4391	0.1501	1.1647	0.2375	0.7312	1.8551	0.6420	0.5209
	seasonwinter*eventhurricane	NA	NA	NA	NA	NA	NA	0.7882	0.1952	0.5376	1.1554	-1.2198	0.2226
myxozoan	(Intercept)	28.1111	0.1598	20.5518	38.4508	20.8767	0.0000	28.4286	0.1619	20.6990	39.0445	20.6768	0.0000
	seasonspring	0.9960	0.6967	0.2543	3.9021	-0.0057	0.9955	0.7309	0.2980	0.4076	1.3107	-1.0521	0.2928
	seasonsummer	0.8699	0.2604	0.5222	1.4492	-0.5351	0.5926	0.7739	0.3459	0.3928	1.5244	-0.7411	0.4586
	seasonwinter	0.9664	0.3199	0.5162	1.8092	-0.1068	0.9149	0.8375	0.2297	0.5340	1.3137	-0.7721	0.4401

eventhurricane	0.6448	0.2356	0.4063	1.0231	-1.8631	0.0625	0.5319	0.2224	0.3439	0.8225	-2.8389	0.0045
seasonspring*eventhurricane	1.0967	0.7812	0.2372	5.0705	0.1182	0.9059	2.1642	0.3678	1.0525	4.4501	2.0992	0.0358
seasonsummer*eventhurricane	1.7366	0.3460	0.8814	3.4213	1.5952	0.1107	2.2256	0.4074	1.0015	4.9458	1.9638	0.0496
seasonwinter*eventhurricane	NA	NA	NA	NA	NA	NA	1.2821	0.3316	0.6694	2.4555	0.7494	0.4536

Supplemental Table 2. Complete ‘*glmmTMB*’ conditional model outputs for model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites, model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites, and model 3 measuring hurricane effects on myxozoan parasites. Bolded values represent significance and bolded model terms represent significant values on both short- and long- time scales.

response variable	reference term	model term	3 month						6 month					
			estimate	std.error	CI_lower	CI_upper	z-statistic	p.value	estimate	CI_lower	CI_upper	std.error	z-statistic	p.value
direct or complex life cycle	complex	(Intercept)	0.3827	0.6618	0.1046	1.4002	-1.4514	0.1467	0.4762	0.3698	0.2307	0.9830	-2.0062	0.0448
		eventhurricane	0.7349	0.5776	0.2369	2.2798	-0.5332	0.5939	1.1808	0.3302	0.6182	2.2557	0.5033	0.6147
		timingpost	1.1577	0.5676	0.3806	3.5218	0.2581	0.7964	0.8683	0.3464	0.4404	1.7120	-0.4078	0.6834
		life_historyDirect	2.6978	0.7143	0.6653	10.9400	1.3894	0.1647	2.2586	0.6360	0.6493	7.8561	1.2811	0.2002
		mill_locationbelow	0.7728	0.2693	0.4558	1.3101	-0.9572	0.3385	1.1227	0.1475	0.8409	1.4989	0.7850	0.4325
		eventhurricane*timingpost	1.1580	0.7558	0.2632	5.0941	0.1941	0.8461	1.3417	0.3988	0.6140	2.9316	0.7371	0.4611
		eventhurricane*life_historyDirect	1.9866	0.4724	0.7871	5.0141	1.4531	0.1462	1.1273	0.4068	0.5079	2.5021	0.2945	0.7684
		timingpost*life_historyDirect	1.2374	0.5142	0.4516	3.3904	0.4143	0.6787	1.3744	0.4471	0.5722	3.3011	0.7113	0.4769
		eventhurricane*timingpost*life_historyDirect	0.1988	0.6764	0.0528	0.7484	-2.3885	0.0169	0.3853	0.5134	0.1409	1.0538	-1.8579	0.0632
	direct	(Intercept)	1.0324	0.8113	0.2105	5.0632	0.0393	0.9687	1.0756	0.5815	0.3441	3.3621	0.1253	0.9003
		eventhurricane	1.4600	0.5812	0.4673	4.5612	0.6511	0.5150	1.3311	0.3336	0.6923	2.5595	0.8574	0.3912
		timingpost	1.4326	0.5566	0.4813	4.2646	0.6460	0.5183	1.1933	0.3712	0.5765	2.4702	0.4762	0.6339
		life_historyComplex	0.3707	0.7143	0.0914	1.5031	-1.3894	0.1647	0.4428	0.6360	0.1273	1.5400	-1.2811	0.2002

monogenean, trematode adult, trematode metacercaria	mill_locationbelow	0.7728	0.2693	0.4558	1.3100	-0.9572	0.3384	1.1227	0.1475	0.8409	1.4989	0.7849	0.4325
	eventhurricane*timingpost	0.2302	0.7861	0.0493	1.0747	-1.8684	0.0617	0.5169	0.4216	0.2262	1.1812	-1.5650	0.1176
	eventhurricane*life_historyComplex	0.5034	0.4724	0.1994	1.2705	-1.4531	0.1462	0.8871	0.4068	0.3997	1.9690	-0.2945	0.7684
	timingpost*life_historyComplex	0.8081	0.5142	0.2950	2.2142	-0.4143	0.6787	0.7276	0.4471	0.3029	1.7476	-0.7114	0.4769
	eventhurricane*timingpost*life_historyComplex	5.0303	0.6764	1.3362	18.9379	2.3885	0.0169	2.5956	0.5134	0.9490	7.0995	1.8580	0.0632
	(Intercept)	2.9847	0.7699	0.6600	13.4978	1.4203	0.1555	1.7263	0.5902	0.5429	5.4892	0.9251	0.3549
	eventhurricane	1.1492	0.5611	0.3827	3.4513	0.2479	0.8042	1.3959	0.3409	0.7157	2.7228	0.9785	0.3278
	timingpost	0.6086	0.5734	0.1978	1.8725	-0.8660	0.3865	1.3419	0.3702	0.6495	2.7725	0.7942	0.4271
	psite_typetrematode	0.3755	0.8500	0.0710	1.9868	-1.1523	0.2492	0.3727	0.6848	0.0974	1.4265	-1.4413	0.1495
	psite_typetrematode_metacercaria	1.3965	0.9986	0.1973	9.8864	0.3345	0.7380	0.9362	0.8739	0.1688	5.1913	-0.0754	0.9399
	mill_locationbelow	1.1619	0.3068	0.6368	2.1199	0.4891	0.6248	1.1506	0.1510	0.8559	1.5469	0.9294	0.3527
	eventhurricane*timingpost	0.5858	0.7800	0.1270	2.7020	-0.6856	0.4930	0.4737	0.4200	0.2080	1.0790	-1.7790	0.0752
	eventhurricane*psite_typetrematode	0.5204	0.5417	0.1800	1.5047	-1.2057	0.2279	1.1439	0.4224	0.4998	2.6179	0.3182	0.7503
	eventhurricane*psite_typetrematode_metacercaria	0.1319	0.7680	0.0293	0.5943	-2.6376	0.0083	0.4220	0.6399	0.1204	1.4793	-1.3481	0.1776
	timingpost*psite_typetrematode	1.6167	0.5472	0.5532	4.7247	0.8779	0.3800	1.1533	0.4676	0.4613	2.8835	0.3050	0.7604
	timingpost*psite_typetrematode_metacercaria	0.1981	0.8263	0.0392	1.0006	-1.9592	0.0501	0.2413	0.6968	0.0616	0.9454	-2.0406	0.0413
	eventhurricane*timingpost*psite_typetrematode	4.4669	0.8142	0.9056	22.0339	1.8382	0.0660	2.1532	0.5470	0.7370	6.2907	1.4022	0.1609
	eventhurricane*timingpost*psite_typetrematode_metacercaria	10.4622	0.9973	1.4814	73.8869	2.3541	0.0186	5.2435	0.8465	0.9979	27.5511	1.9576	0.0503
	(Intercept)	4.1682	0.7861	0.8930	19.4563	1.8160	0.0694	1.5894	0.6625	0.4338	5.8233	0.6994	0.4843
	eventhurricane	0.1516	0.7686	0.0336	0.6839	-2.4544	0.0141	0.6007	0.5525	0.2034	1.7740	-0.9224	0.3563
timingpost	0.1206	0.8041	0.0249	0.5831	-2.6309	0.0085	0.3362	0.5845	0.1069	1.0572	-1.8648	0.0622	
psite_typetrematode	0.2689	0.8827	0.0477	1.5169	-1.4880	0.1368	0.3954	0.7386	0.0930	1.6817	-1.2562	0.2091	
psite_typemonogenean	0.7161	0.9986	0.1011	5.0694	-0.3345	0.7380	1.1194	0.8218	0.2236	5.6037	0.1372	0.8908	
mill_locationbelow	1.1619	0.3068	0.6368	2.1199	0.4891	0.6248	1.1591	0.1525	0.8595	1.5630	0.9678	0.3332	
eventhurricane*timingpost	6.1289	0.9966	0.8690	43.2254	1.8191	0.0689	2.3013	0.7074	0.5752	9.2070	1.1782	0.2387	
eventhurricane*psite_typetrematode	3.9451	0.7751	0.8636	18.0214	1.7708	0.0766	2.6818	0.6002	0.8271	8.6957	1.6437	0.1002	
eventhurricane*psite_typemonogenean	7.5803	0.7680	1.6826	34.1494	2.6376	0.0083	2.2778	0.5852	0.7234	7.1724	1.4066	0.1595	
timingpost*psite_typetrematode	8.1598	0.7871	1.7445	38.1672	2.6670	0.0077	4.7628	0.6405	1.3573	16.7128	2.4370	0.0148	
timingpost*psite_typemonogenean	5.0473	0.8263	0.9994	25.4915	1.9592	0.0501	3.8808	0.6398	1.1074	13.5998	2.1195	0.0341	

		eventhurricane*timingpost*psite_typetrematode	0.4270	1.0282	0.0569	3.2036	-0.8277	0.4078	0.4301	0.7774	0.0937	1.9739	-1.0853	0.2778
		eventhurricane*timingpost*psite_typemonogenean	0.0956	0.9973	0.0135	0.6750	-2.3541	0.0186	0.2159	0.7641	0.0483	0.9655	-2.0060	0.0449
		(Intercept)	0.1259	0.8654	0.0231	0.6867	-2.3944	0.0166	0.0617	0.7475	0.0143	0.2670	-3.7266	0.0002
		eventhurricane	0.3158	0.9239	0.0516	1.9309	-1.2478	0.2121	0.4634	0.6242	0.1363	1.5752	-1.2321	0.2179
myxozoan	(no triple interaction)	timingpost	0.6002	0.7133	0.1483	2.4292	-0.7157	0.4742	1.2768	0.6078	0.3879	4.2027	0.4020	0.6877
		mill_locationbelow	0.8125	0.4848	0.3141	2.1015	-0.4283	0.6685	1.3015	0.3463	0.6602	2.5659	0.7610	0.4467
		eventhurricane*timingpost	0.9773	1.2293	0.0878	10.8755	-0.0187	0.9851	0.6187	0.7976	0.1296	2.9544	-0.6019	0.5473

Supplemental Table 3. Outputs of ‘*glmmTMB*’ zero-inflated model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites, model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites, and model 3 measuring hurricane effects on myxozoan parasites. Bolded values represent significance and bolded model terms represent significant values on both short- and long- time scales.

response variable	reference term	model term	3 month						6 month					
			estimate	std.error	CI_lower	CI_upper	z-statistic	p.value	estimate	std.error	CI_lower	CI_upper	z-statistic	p.value
		(Intercept)	29.1632	0.5275	10.3710	82.0065	6.3942	0.0000	16.2467	0.4595	6.6019	39.9817	6.0678	0.0000
		eventhurricane	0.6156	0.4616	0.2491	1.5211	-1.0513	0.2931	1.1033	0.4402	0.4656	2.6144	0.2233	0.8233
		timingpost	0.8206	0.4971	0.3097	2.1740	-0.3978	0.6908	0.9394	0.4566	0.3839	2.2989	-0.1369	0.8911
direct or complex life cycle strategy	complex	life_historyDirect	0.1132	0.8638	0.0208	0.6153	-2.5221	0.0117	0.2082	0.8052	0.0430	1.0089	-1.9490	0.0513
		mill_locationbelow	0.8806	0.2743	0.5144	1.5077	-0.4633	0.6431	1.0964	0.1862	0.7612	1.5794	0.4944	0.6210
		eventhurricane*timingpost	2.5932	0.6295	0.7551	8.9056	1.5138	0.1301	1.7425	0.5335	0.6125	4.9575	1.0410	0.2979
		eventhurricane*life_historyDirect	2.0153	0.6977	0.5133	7.9118	1.0043	0.3152	1.8563	0.6485	0.5207	6.6176	0.9538	0.3402

		timingpost*life_historyDirect	5.9783	0.7002	1.5156	23.5811	2.5539	0.0107	4.9312	0.6718	1.3215	18.4001	2.3750	0.0175
		eventhurricane*timingpost*life_historyDirect	0.1633	1.0320	0.0216	1.2345	-1.7559	0.0791	0.2519	0.7676	0.0560	1.1340	-1.7962	0.0725
		(Intercept)	3.3012	0.8249	0.6554	16.6288	1.4478	0.1477	3.3822	0.7609	0.7611	15.0290	1.6013	0.1093
		eventhurricane	1.2405	0.6017	0.3815	4.0341	0.3583	0.7201	2.0480	0.6135	0.6153	6.8167	1.1685	0.2426
		timingpost	4.9057	0.6613	1.3422	17.9301	2.4051	0.0162	4.6323	0.6240	1.3635	15.7374	2.4570	0.0140
		life_historyComplex	8.8338	0.8638	1.6251	48.0197	2.5221	0.0117	4.8036	0.8052	0.9911	23.2802	1.9490	0.0513
monogenean, trematode adult, trematode metacercaria	direct	mill_locationbelow	0.8806	0.2743	0.5144	1.5077	-0.4633	0.6431	1.0964	0.1862	0.7612	1.5794	0.4944	0.6210
		eventhurricane*timingpost	0.4235	0.9784	0.0622	2.8821	-0.8781	0.3799	0.4389	0.7228	0.1064	1.8101	-1.1391	0.2547
		eventhurricane*life_historyComplex	0.4962	0.6977	0.1264	1.9480	-1.0043	0.3152	0.5387	0.6485	0.1511	1.9204	-0.9538	0.3402
		timingpost*life_historyComplex	0.1673	0.7002	0.0424	0.6598	-2.5539	0.0107	0.2028	0.6718	0.0543	0.7567	-2.3750	0.0175
		eventhurricane*timingpost*life_historyComplex	6.1231	1.0320	0.8101	46.2829	1.7559	0.0791	3.9699	0.7676	0.8819	17.8706	1.7962	0.0725
		(Intercept)	1.7313	1.3939	0.1127	26.6012	0.3938	0.6938	1.1278	1.0203	0.1527	8.3312	0.1179	0.9062
		eventhurricane	1.3164	0.7932	0.2781	6.2317	0.3465	0.7289	2.6108	0.7510	0.5991	11.3771	1.2779	0.2013
		timingpost	2.4514	1.0412	0.3185	18.8649	0.8612	0.3891	6.0584	0.7522	1.3871	26.4611	2.3950	0.0166
		psite_typedtrematode	8.6200	1.6447	0.3431	216.5392	1.3097	0.1903	8.2139	1.1627	0.8411	80.2113	1.8112	0.0701
		psite_typedtrematode_metacercaria	20.0054	1.6209	0.8345	479.5968	1.8484	0.0645	29.5698	1.2466	2.5690	340.3608	2.7169	0.0066
		mill_locationbelow	1.0892	0.3783	0.5189	2.2861	0.2259	0.8213	1.2881	0.2265	0.8264	2.0080	1.1179	0.2636
		eventhurricane*timingpost	0.4443	1.2843	0.0358	5.5069	-0.6316	0.5276	0.3302	0.8387	0.0638	1.7089	-1.3211	0.1866

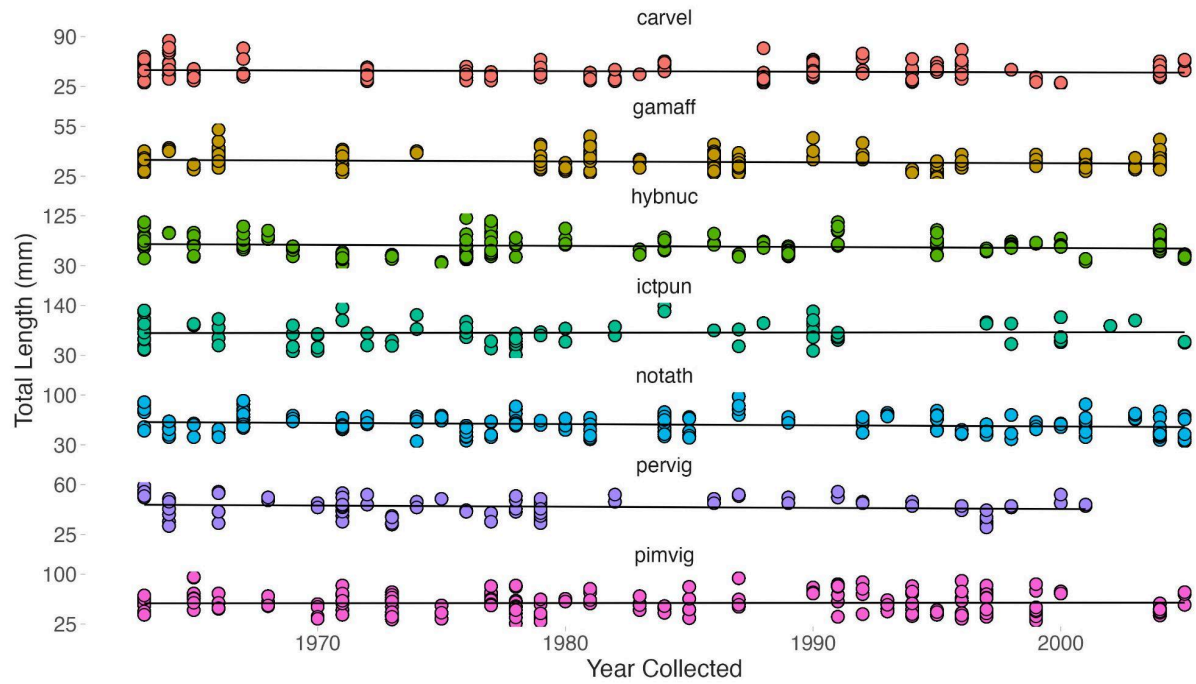
	eventhurricane*psite_typetrematode	0.6698	1.0188	0.0909	4.9331	-0.3934	0.6940	1.0532	0.8659	0.1930	5.7484	0.0599	0.9523
	eventhurricane*psite_typetrematode_metacercaria	0.1326	1.0771	0.0161	1.0945	-1.8762	0.0606	0.0943	0.9469	0.0147	0.6032	-2.4937	0.0126
	timingpost*psite_typetrematode	0.2492	0.9567	0.0382	1.6251	-1.4525	0.1464	0.3452	0.8838	0.0611	1.9518	-1.2033	0.2289
	timingpost*psite_typetrematode_metacercaria	0.0691	1.1264	0.0076	0.6282	-2.3727	0.0177	0.0439	1.0146	0.0060	0.3207	-3.0810	0.0021
	eventhurricane*timingpost*psite_typetrematode	4.8219	1.4954	0.2572	90.3959	1.0520	0.2928	1.9959	0.9704	0.2979	13.3702	0.7122	0.4764
	eventhurricane*timingpost*psite_typetrematode_metacercaria	9.5490	1.5637	0.4456	204.6356	1.4430	0.1490	20.1547	1.1905	1.9545	207.8353	2.5229	0.0116
	(Intercept)	34.6363	1.1223	3.8393	312.4767	3.1587	0.0016	33.8846	0.8453	6.4631	177.6496	4.1675	0.0000
	eventhurricane	0.1745	0.8040	0.0361	0.8436	-2.1715	0.0299	0.2733	0.6686	0.0737	1.0135	-1.9399	0.0524
	timingpost	0.1693	1.1039	0.0195	1.4737	-1.6087	0.1077	0.2754	0.7173	0.0675	1.1235	-1.7977	0.0722
	psite_typetrematode	0.4309	1.3687	0.0295	6.3015	-0.6151	0.5385	0.2791	1.0249	0.0374	2.0801	-1.2454	0.2130
	psite_typemonogenean	0.0500	1.6209	0.0021	1.1983	-1.8484	0.0645	0.0339	1.2395	0.0030	0.3843	-2.7315	0.0063
	mill_locationbelow	1.0892	0.3783	0.5189	2.2861	0.2259	0.8213	1.2568	0.2354	0.7924	1.9935	0.9712	0.3314
trematode													
metacercaria	eventhurricane*timingpost	4.2428	1.2244	0.3850	46.7596	1.1804	0.2379	5.8587	0.8810	1.0420	32.9422	2.0067	0.0448
	eventhurricane*psite_typetrematode	5.0530	1.0248	0.6780	37.6607	1.5807	0.1139	10.2124	0.8191	2.0505	50.8631	2.8366	0.0046
	eventhurricane*psite_typemonogenean	7.5439	1.0771	0.9137	62.2874	1.8762	0.0606	9.2632	0.8885	1.6236	52.8499	2.5055	0.0122
	timingpost*psite_typetrematode	3.6071	0.9823	0.5260	24.7361	1.3060	0.1916	7.3165	0.8817	1.2996	41.1908	2.2572	0.0240
	timingpost*psite_typemonogenean	14.4760	1.1263	1.5918	131.6489	2.3727	0.0177	19.1966	0.9411	3.0347	121.4312	3.1396	0.0017
	eventhurricane*timingpost*psite_typetrematode	0.5050	1.4465	0.0296	8.6014	-0.4723	0.6367	0.1168	1.0416	0.0152	0.8995	-2.0617	0.0392

		eventhurricane*timingpost*psite_typemonogenean	0.1047	1.5637	0.0049	2.2443	-1.4430	0.1490	0.0681	1.1054	0.0078	0.5946	-2.4304	0.0151
		(Intercept)	1.2664	0.7552	0.2882	5.5636	0.3127	0.7545	0.4223	0.6312	0.1225	1.4553	-1.3656	0.1721
		eventhurricane	0.6705	0.7028	0.1691	2.6587	-0.5687	0.5696	1.2552	0.5468	0.4298	3.6658	0.4156	0.6777
myxozoan	(no triple interaction)	timingpost	0.2961	0.5944	0.0924	0.9491	-2.0478	0.0406	1.1008	0.5340	0.3865	3.1349	0.1798	0.8573
		mill_locationbelow	0.8961	0.3569	0.4452	1.8038	-0.3073	0.7586	1.7798	0.2846	1.0188	3.1092	2.0255	0.0428
		eventhurricane*timingpost	5.8984	0.9152	0.9811	35.4633	1.9391	0.0525	0.8636	0.6874	0.2245	3.3223	-0.2134	0.8310

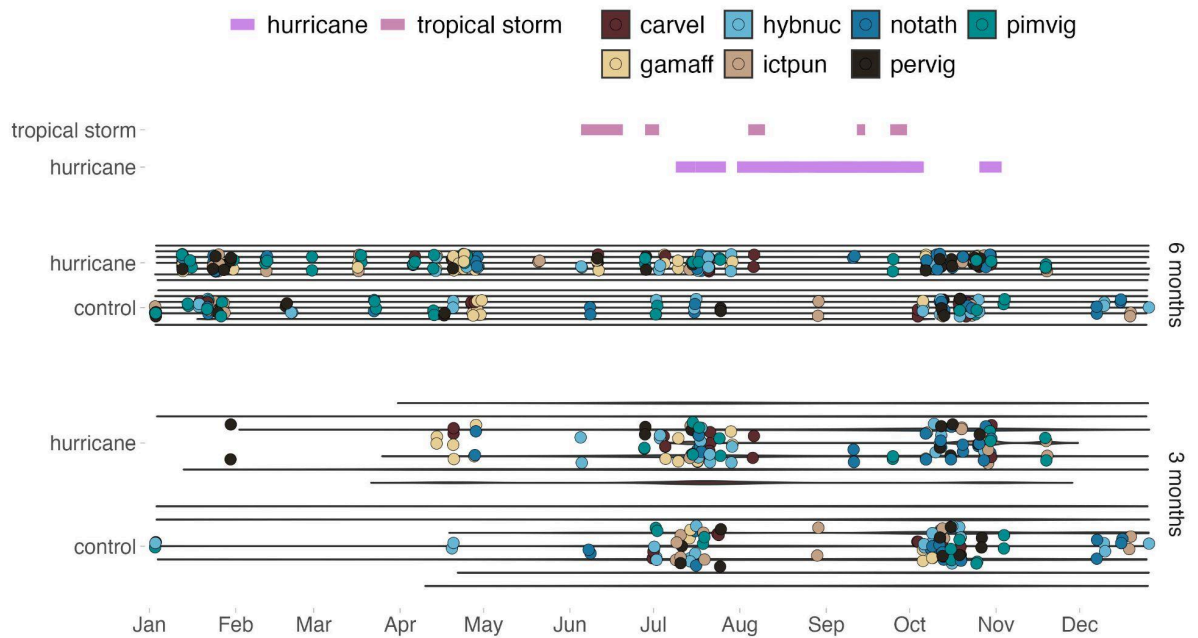
Supplemental Table 4. Outputs of ‘*emmeans*’ ‘*joint_tests*’ function for model 1 measuring hurricane effects on transmission strategy between direct or complex life cycle parasites and model 2 measuring hurricane effects on trematode adult, trematode metacercaria, and monogenean parasites.

response variable	reference term	model term	3 month				6 month			
			df	χ ²	F.ratio	p.value	df	χ ²	F.ratio	p.value
direct or complex life cycle	complex	event	1.0000	0.6770	0.6770	0.4107	1.0000	0.7030	0.7030	0.4019
		timing	1.0000	0.0580	0.0580	0.8104	1.0000	0.1760	0.1760	0.6749
		life_history	1.0000	2.8390	2.8390	0.0920	1.0000	2.2620	2.2620	0.1326
		ci	1.0000	0.9160	0.9160	0.3385	1.0000	0.6160	0.6160	0.4325
		event:timing	1.0000	0.9100	0.9100	0.3401	1.0000	0.3270	0.3270	0.5677
		event:life_history	1.0000	0.1380	0.1380	0.7102	1.0000	2.2190	2.2190	0.1363
		timing:life_history	1.0000	2.8450	2.8450	0.0917	1.0000	0.3340	0.3340	0.5631
		event:timing:life_history	1.0000	5.7050	5.7050	0.0169	1.0000	3.4520	3.4520	0.0632
	direct	event	1.0000	0.6770	0.6770	0.4107	1.0000	0.7030	0.7030	0.4019
		timing	1.0000	0.0580	0.0580	0.8104	1.0000	0.1760	0.1760	0.6749
		life_history	1.0000	2.8390	2.8390	0.0920	1.0000	2.2620	2.2620	0.1326

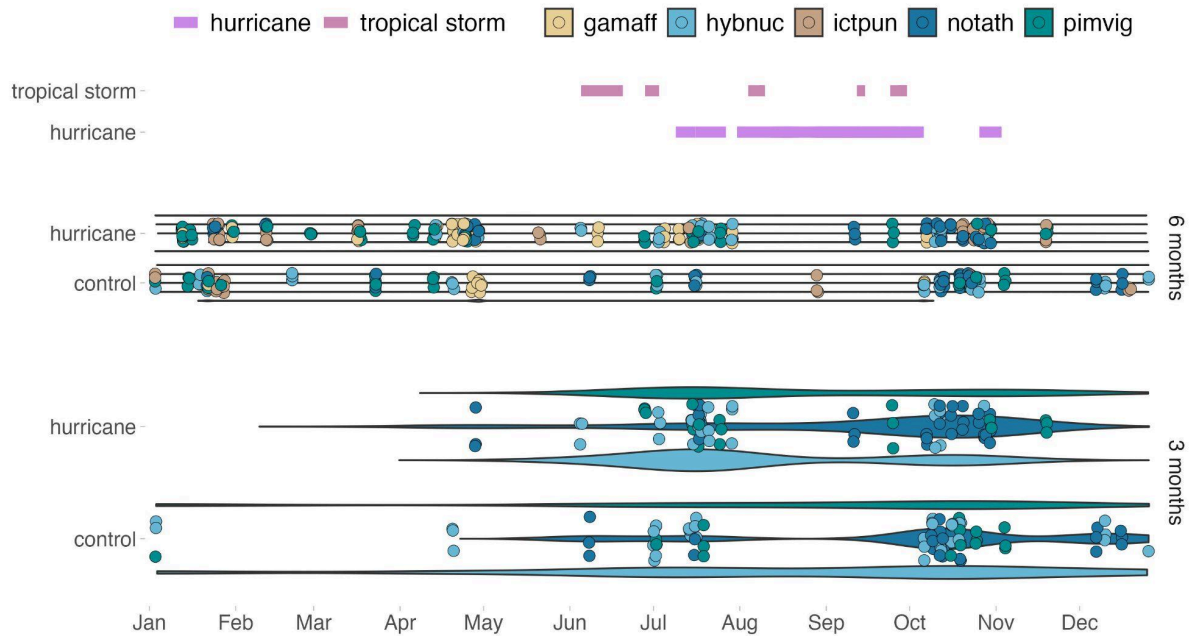
monogenean, trematode adult, trematode metacercaria	monogenean	ci	1.0000	0.9160	0.9160	0.3384	1.0000	0.6160	0.6160	0.4325
		event:timing	1.0000	0.9100	0.9100	0.3401	1.0000	0.3270	0.3270	0.5677
		event:life_history	1.0000	0.1380	0.1380	0.7102	1.0000	2.2190	2.2190	0.1363
		timing:life_history	1.0000	2.8450	2.8450	0.0917	1.0000	0.3340	0.3340	0.5631
		event:timing:life_history	1.0000	5.7050	5.7050	0.0169	1.0000	3.4520	3.4520	0.0632
		event	1.0000	1.0710	1.0710	0.3008	1.0000	0.4270	0.4270	0.5135
		timing	1.0000	2.4980	2.4980	0.1140	1.0000	0.2990	0.2990	0.5843
		psite_type	2.0000	1.3780	0.6890	0.5023	2.0000	1.6680	0.8340	0.4345
		ci	1.0000	0.2390	0.2390	0.6248	1.0000	0.8640	0.8640	0.3527
		event:timing	1.0000	1.3540	1.3540	0.2445	1.0000	0.0260	0.0260	0.8717
	trematode metacercaria	event:psite_type	2.0000	3.4020	1.7010	0.1825	2.0000	4.7020	2.3510	0.0953
		timing:psite_type	2.0000	13.6420	6.8210	0.0011	2.0000	8.4360	4.2180	0.0147
		event:timing:psite_type	2.0000	6.6260	3.3130	0.0364	2.0000	4.5580	2.2790	0.1024
		event	1.0000	1.0710	1.0710	0.3008	1.0000	0.3620	0.3620	0.5473
		timing	1.0000	2.4980	2.4980	0.1140	1.0000	0.2730	0.2730	0.6015
		psite_type	2.0000	1.3780	0.6890	0.5023	2.0000	1.8280	0.9140	0.4007
		ci	1.0000	0.2390	0.2390	0.6248	1.0000	0.9370	0.9370	0.3332
		event:timing	1.0000	1.3540	1.3540	0.2445	1.0000	0.0130	0.0130	0.9091
		event:psite_type	2.0000	3.4020	1.7010	0.1825	2.0000	4.6980	2.3490	0.0955
		timing:psite_type	2.0000	13.6420	6.8210	0.0011	2.0000	9.1540	4.5770	0.0103
		event:timing:psite_type	2.0000	6.6260	3.3130	0.0364	2.0000	4.4040	2.2020	0.1106



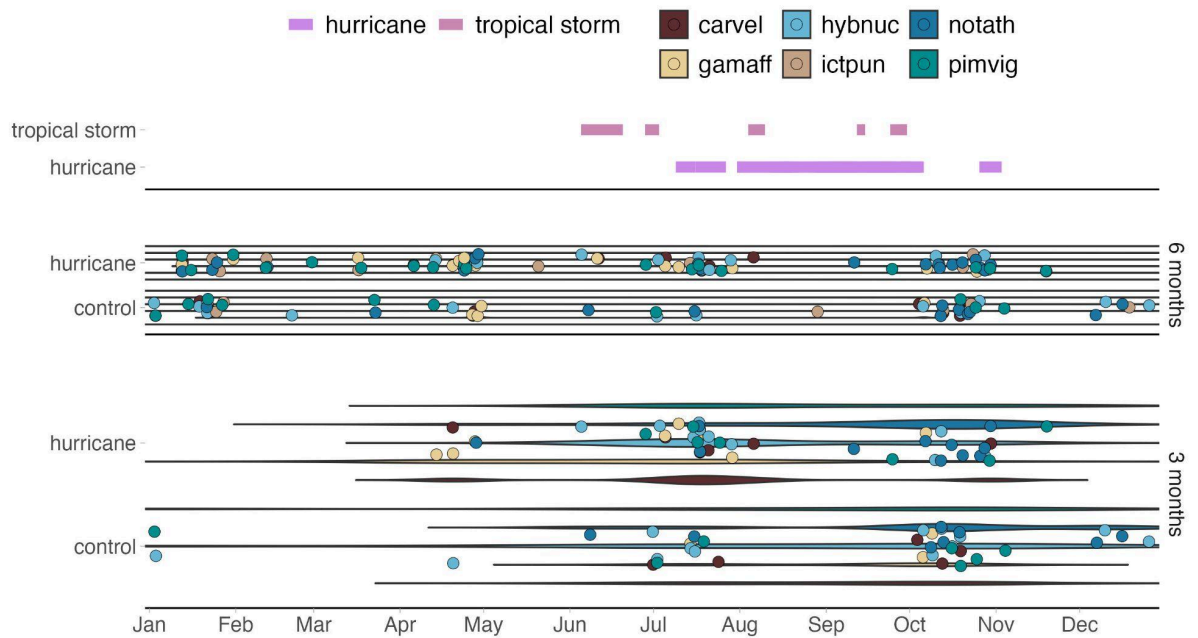
Supplemental Figure 9. Fish body size (total length, in mm) through time. Each facet represents one fish species (gamaff = *Gambusia affinis*, carvel = *Carpiodes velifer*, hybnuc = *Hybognathus nuchalis*, ictpun = *Ictalurus punctatus*, notath = *Notropis atherinoides*, pervig = *Percina vigil*, and pimvig = *Pimephales vigilax*) and black lines are linear regression lines. The flat slope of the line indicates insignificant relationships between total length and time across all fish species measured.



Supplemental Figure 10. Effect of season of collection on transmission strategy analysis. Datapoints were assigned an arbitrary year of occurrence to their accurate day/month (in this case 2025) in order to map collection across one year to assess seasonality. Width of bars represents higher counts of fish lots and raw data are depicted as dots where color corresponds to fish species. The colored bars at the top represent when tropical storm and hurricane events occurred in the assigned representative year.



Supplemental Figure 11. Effect of season of collection on parasite taxonomic group analysis (monogeneans, trematode adults, and trematode metacercaria). Datapoints were assigned an arbitrary year of occurrence to their accurate day/month (in this case 2025) in order to map collection across one year to assess seasonality. Width of bars represents higher counts of fish lots and raw data are depicted as dots where color corresponds to fish species. The colored bars at the top represent when tropical storm and hurricane events occurred in the assigned representative year.



Supplemental Figure 12. Effect of season of collection on myxozoan abundance analysis. Datapoints were assigned an arbitrary year of occurrence to their accurate day/month (in this case 2025) in order to map collection across one year to assess seasonality. Width of bars represents higher counts of fish lots and raw data are depicted as dots where color corresponds to fish species. The colored bars at the top represent when tropical storm and hurricane events occurred in the assigned representative year.